

**Problem 1.** [Total 6-points] For each part simplify your final answer as much as possible.

|         | $X = 0$ | $X = 1$ | $X = 2$ |
|---------|---------|---------|---------|
| $Y = 1$ | 0.15    | 0.1     | 0       |
| $Y = 0$ | 0.25    | 0.3     | 0.2     |

(a) Find  $P(Y = 1 | X \leq 1)$ .

(b) Find  $\text{Var}(X | Y = 1)$ .

(c) Find  $E[16Y - 4X | X \leq 1]$

$$\begin{aligned}
 P(Y=1 | X \leq 1) &= \frac{P(X=0, Y=1) + P(X=1, Y=1)}{P(X \leq 1)} \\
 &= \frac{P(X=0, Y=1) + P(X=1, Y=1)}{1 - P(X=2)} \\
 &= \frac{0.15 + 0.1}{0.8} = \frac{0.25}{0.8} = \frac{25}{80} = \frac{5}{16}
 \end{aligned}$$

$$(b) \quad X | Y=1 \sim \text{Bern}(p = P(X=1 | Y=1))$$

$$P(X=1 | Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{0.1}{0.25} = \frac{10}{25} = \frac{2}{5}$$

$$\text{Var}(X | Y=1) = p(1-p) = \frac{2}{5} \left( \frac{3}{5} \right) = \frac{6}{25}$$

$$(c) \quad Y | X \leq 1 \sim \text{Bern}(p_1 = P(Y=1 | X \leq 1))$$

$$X | X \leq 1 \sim \text{Bern}(p_2 = P(X=1 | X \leq 1))$$

$$\text{From (a)} \quad p_1 = \frac{5}{16}$$

$$p_2 = P(X=1 | X \leq 1) = \frac{P(X=1)}{P(X \leq 1)} = \frac{0.1 + 0.3}{0.8} = \frac{0.4}{0.8} = \frac{1}{2}$$

By linearity of expectation

$$\begin{aligned} E[16Y - 4X | X \leq 1] &= 16 \underbrace{E[Y | X \leq 1]}_{\downarrow} - 4 \underbrace{E[X | X \leq 1]}_{\downarrow} \\ &= 16 \underbrace{P(Y=1 | X \leq 1)}_{\text{green}} - 4 \underbrace{P(X=1 | X \leq 1)}_{\text{red}} \\ &= 16 \cdot \frac{5}{16} - 4 \cdot \frac{1}{2} = 5 - 2 = 3 \end{aligned}$$

**Problem 2.** [Total 8 points] Show all your work.Let the joint probability density function for  $X$  and  $Y$  be given as

$$f(x, y) = \begin{cases} Cx(1-x)e^{-2y} & 0 < x < 1, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the constant  $C$  to make the above a valid pdf.  
 (b) Find  $\text{Var}(4Y+3)$ .  
 (c) Find  $\text{Cov}(X, Y)$ .

$$\int_0^{\infty} \int_0^1 C x(1-x) e^{-2y} dx dy = 1$$

$$\int_0^1 \overbrace{x(1-x)}^{x-x^2} dx \int_0^{\infty} e^{-2y} dy = \frac{1}{C}$$

$$\left( \frac{x^2}{2} - \frac{x^3}{3} \right) \bigg|_0^1 \cdot \left( -\frac{e^{-2y}}{2} \right) \bigg|_0^{\infty} = \frac{1}{C}$$

$$\left( \frac{1}{2} - \frac{1}{3} \right) \left( 0 + \frac{1}{2} \right) = \frac{1}{C}$$

$$\Rightarrow \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{C} \Rightarrow C = 12$$

(b) First find pdf of  $Y$ 

$$f(y) = \begin{cases} \int_0^1 12x(1-x) e^{-2y} dx & 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

$$f(y) = \begin{cases} 12e^{-2y} \int_0^1 x(1-x) dx & 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

in part (a)  
we saw this  
part 2 is  $\frac{1}{6}$

$$f(y) = \begin{cases} 2e^{-2y} & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow Y \sim \text{Exp}(\lambda=2) \Rightarrow \text{Var}(Y) = \frac{1}{\lambda^2} = \frac{1}{4}$$

$$\text{Var}(4Y+3) = 16 \text{Var}(Y) = 16 \cdot \frac{1}{4} = 4$$

$$(c) \quad f(x, y) = (\text{A function of } x) \text{ times } (\text{A function of } y)$$

$$\Rightarrow X, Y \text{ are independent}$$

$$\Rightarrow \text{Cov}(X, Y) = 0$$



**Problem 3.** [8 points] There are 20 students in a classroom. Each student hands in their homework in a folder with their name on it. The teacher shuffles the folders and hands them back randomly, one to each student.

Let  $X$  be number of students who get back their own homework folder. Calculate  $\text{Var}(X)$ .  
SHOW ALL YOUR WORK.

$$X = X_1 + X_2 + \dots + X_n \quad \text{where}$$

$$X_i = \begin{cases} 1 & \text{if the student } i \text{ picks} \\ & \text{their own name} \\ 0 & \text{otherwise} \end{cases}$$

and  $n = 20$ .

each  $X_i \sim \text{Bern}(p = P(X_i = 1))$

$$P(X_i = 1) = \frac{1}{n} \Rightarrow \text{Var}(X_i) = \frac{1}{n} \left(1 - \frac{1}{n}\right)$$

$$\text{Var}(X) = \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j\right) = \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)$$

Simplification  
by matrix

$$X_i X_j = \begin{cases} 1 & \text{if } X_i = 1 \text{ and } X_j = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$X_i X_j \sim \text{Bern}(p = P(X_i = 1 \text{ and } X_j = 1))$$

$$P(X_i = 1 \text{ and } X_j = 1) = P(X_1 = 1 \text{ and } X_2 = 1) = P(X_2 = 1 \mid X_1 = 1) P(X_1 = 1) = \frac{1}{n-1} \cdot \frac{1}{n}$$

$$E[X_i X_j] = \frac{1}{n-1} \cdot \frac{1}{n}$$

$$\text{Var}(X) = \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j\right) \underset{\uparrow}{=} \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)$$

$$\text{Var}(X) = n \text{Var}(X_1) + n(n-1) \left( E[X_1 X_2] - E[X_1] E[X_2] \right)$$

$$= n \frac{1}{n} \left( 1 - \frac{1}{n} \right) + n(n-1) \left[ \frac{1}{n-1} \cdot \frac{1}{n} - \frac{1}{n^2} \right]$$

$$= 1 - \frac{1}{n} + (n-1) \left[ \frac{1}{n-1} - \frac{1}{n} \right]$$

$$= 1 - \frac{1}{n} + \frac{n - n + 1}{n(n-1)} (n-1)$$

$$= 1 - \frac{1}{n} + \frac{1}{n} = 1$$

$$\Rightarrow \boxed{\text{Var}(X) = 1}$$

**Problem 4.** The followings are for Part(a) and (b)

$$\begin{aligned} E[X] = \mu_X = 1, \quad \text{Var}(X) = \sigma_X^2 = 4, \quad X \sim N(\mu_X, \sigma_X^2) \\ E[Y] = \mu_Y = -2, \quad \text{Var}(Y) = \sigma_Y^2 = 9 \end{aligned}$$

Evaluate the followings. SHOW ALL YOUR WORK.

(a)  $\text{Var}(3X + \frac{Y}{3})$

(b) Find the value of Cdf of X at  $X = 5$ , meaning  $F_X(5)$

(c) Find  $P(2 < |X - 1| < 4)$

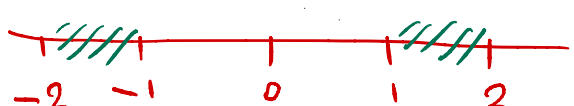
$$\begin{aligned} \text{(a)} \quad \text{Var}(3X + \frac{Y}{3}) &= \text{Cov}(3X + \frac{Y}{3}, 3X + \frac{Y}{3}) \\ &= 9 \text{Var}(X) + \frac{\text{Var}(Y)}{9} + 2 \text{Cov}(X, Y) \\ &= 36 + 1 + 2 \text{Cov}(X, Y) = 37 + 2 \text{Cov}(X, Y) \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad F_X(5) &= P(X \leq 5) = P\left(\frac{X - \mu_X}{\sqrt{\text{Var}(X)}} \leq \frac{5 - 1}{2}\right) = P(Z \leq 2) \\ &= \Phi(2) \end{aligned}$$

By 68 - 95 - 99.7

$$P(|Z| < 2) = 0.95 \Rightarrow P(0 < Z < 2) = \frac{0.95}{2}$$

$$\Rightarrow \Phi(2) - \Phi(0) = \frac{0.95}{2} \quad \Phi(0) = \frac{1}{2} \Rightarrow \Phi(2) = \frac{1.95}{2}$$

$$\begin{aligned} \text{(c)} \quad P(1 < \frac{|X - 1|}{2} < 2) &= P(1 < |Z| < 2) = P(1 < Z < 2) + P(-2 < Z < -1) \\ &= 0.95 - 0.68 = 0.27 \end{aligned}$$


**Problem 5.** [6 points] Let  $T$  denote the time it takes for a computer to shut down. Suppose  $T$  follows an Exponential distribution with mean 0.5 minutes. A computer lab has 6 independent computers that must all be shut down at the end of the day.

What is the probability that at most 2 computers take longer than 1 minute to shut down on a given day?

$$T \sim \text{Exp}(\lambda) \Rightarrow E[T] = \frac{1}{\lambda}$$

$$E[T] = 0.5 \Rightarrow \boxed{\lambda = 2}$$

$$P(T > 1) = \int_1^{\infty} \lambda e^{-\lambda t} dt = -e^{-\lambda t} \Big|_1^{\infty} = e^{-\lambda}$$

$$X \sim \text{Bin}(n=6, p=P(T > 1))$$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= (1 - e^{-\lambda})^6 + \binom{6}{1} e^{-\lambda} (1 - e^{-\lambda})^5 + \binom{6}{2} e^{-2\lambda} (1 - e^{-\lambda})^4$$

$$= (1 - e^{-2})^4 \left[ (1 - e^{-2})^4 + \binom{6}{1} e^{-2} (1 - e^{-2}) + \binom{6}{2} e^{-4} \right]$$

**Problem 6.** [6 points] A factory produces light bulbs. This factory has two branches Branch A and Branch B. The probability of a bulb being defective if it is made in Branch A is 20%, while the probability of a bulb being defective if it is made in Branch B is 5%. Assume 30% of the products are made in Branch A.

A random sample of 10 bulbs was inspected, and 3 of them were found to be defective. What is the probability that these bulbs had been made in Branch A.

$D$  = being defective

$$P(D|A) = 0.2, \quad P(D|B) = 0.05, \quad P(A) = 0.3$$

$X$  = number of defective bulbs out of 10 bulbs

$$P(A|X=3) = \frac{P(X=3|A)P(A)}{P(X=3)}$$

$P$

$$X \sim \text{Bin}(n=10, p=P(D))$$

$$X|A \sim \text{Bin}(n=10, p=P(D|A))$$

$$P(A|X=3) = \frac{\binom{10}{3} (0.2)^3 (1-0.2)^7 \cdot 0.3}{\binom{10}{3} (0.05)^3 (1-0.05)^7} = \frac{(0.2)^3 (0.8)^7 (0.3)}{(0.05)^3 (0.95)^7}$$

**Problem 7.** Sarah works as a sales representative selling house insurance over the phone. Only 10% of all calls result in a successful sale. Assume that the outcome of each call is independent of the others.

For each part only state the name of distribution and its parameters and set up your probability no need to evaluate anything, for example write: " $X \sim \text{Bin}(n = 10, p = 0.2)$ " and my answer to this question is  $P(X \geq 2)$ "

- [1 point] What is the probability that Sarah makes her second sale on the 9<sup>th</sup> call?
- [1 point] What is the probability that Sarah makes her first sale on the third call?
- [1 point] What is the probability that out of 20 call Sarah makes at least 3 sales?

$$(a) \quad X \sim \text{NBin}(r=2, p=0.1) \quad , \quad P(X=9)$$

$$(b) \quad Y \sim \text{Geom}(p=0.1) \quad , \quad P(Y=3)$$

$$(c) \quad W \sim \text{Bin}(n=20, p=0.1) \quad , \quad P(W \geq 3)$$

**Problem 8.** A coffee shop receives, on average, 10 customers per hour.

For each part only state the name of distribution and its parameters and set up your probability no need to evaluate anything.

- [1 point] What is the probability that exactly 7 customers arrive in a given hour?
- [1 point] What is the probability that the time between two consecutive customer arrivals is more than 10 minutes?
- [1 point] Suppose the shop is open for 30 hours, and find the probability that there are exactly 2 hours with no customers at all.

$$(a) \quad X \sim \text{Pois}(\lambda=10) \quad , \quad P(X=7)$$

$$(b) \quad T \sim \text{Exp}(\lambda=10) \quad , \quad 10 \text{ minute} = \frac{1}{6} \text{ hour} \quad , \quad P(T > \frac{1}{6})$$

$$(c) \quad Y \sim \text{Bin}(n=30, p=P(X=0)) \quad , \quad p=P(X=0)=e^{-10}, \quad P(Y=2)$$

**Problem 9.** For part (a) and (b) determine whether it is true or false

- [0.5 point]  $\text{Var}(2X-Y) = 4\text{Var}(X) + \text{Var}(Y)$ . This means X and Y are independent. **F**
- [0.5 point]  $Y = 2X$ , then  $\text{Cov}(Y, X)$  must be positive. **F**
- [0.5 point] We know as X is decreasing Y will increase, what is sign of  $\text{Cov}(X, Y)$ . **negative**

$$(a) \quad \text{var}(2X - Y) = 4\text{var}(X) + \text{var}(Y) \text{ only means } \text{Cov}(X, Y) = 0$$

$$(b) \quad \text{Cov}(2X, X) = 2\text{var}(X) \text{ but } \text{var}(X) \geq 0, \text{ if } X \text{ is constant } \text{var}(X) = 0$$